

Stat 220: A Map of Models

Midterm practice questions, with answers

Practice on what separates a probability model from a non-probability one, what each lets you ask, and how the job you were given narrows the choice. No computer needed.

1. What makes something a probability model?

- A. It states a distribution the outcome is drawn from.
- B. It uses probability somewhere in the fitting.
- C. Its predictions fall between 0 and 1.
- D. It was fit by maximum likelihood.

Answer: A. A probability model commits to where the data came from, saying the outcome is normal around a line, or Bernoulli with some probability, or Poisson with some rate. That commitment is what produces a likelihood, and the likelihood is what standard errors, p-values, confidence intervals and AIC are all made of. The last option names a fitting method, which comes after the distribution rather than instead of it.

2. A colleague wants the p-value on the top variable in a random forest. What do you say?

- A. It is in the output, you have to request it.
- B. Bootstrap the forest and read it off.
- C. It is one over the number of trees.
- D. There isn't one to report.

Answer: D. A forest never says what distribution the data came from, so there is no likelihood, and a p-value has nothing to be computed from. The quantity does not exist rather than being hidden. Resampling will tell you how much an importance score wobbles, which is worth knowing, but that is a stability check and not a test of anything.

3. You want to compare a linear regression against a boosted model. Which comparison works?

- A. Adjusted R-squared on the training data.
- B. Cross-validated error, same folds for both.
- C. AIC, which exists to compare models.
- D. Neither, the two are different kinds of model.

Answer: B. AIC needs a likelihood, so boosting is not eligible. Out-of-sample error works on anything that can produce a prediction, which is why it is the general-purpose answer. The catch is in the second half of the option: both models have to face identical folds, or you are partly measuring which one drew the easier split.

4. Which of these does fitting decide, rather than you?

- A. How deep a tree may grow.
- B. How many folds to cross-validate with.
- C. Where a tree puts its splits.
- D. How big a penalty to apply.

Answer: C. Fitting settles the parts the data has an opinion about: split points, slopes, weights. Depth, fold count and penalty size are settings you fix beforehand, and no amount of fitting will choose them for you. Unit 4 covers how to choose them without fooling yourself.

5. Your outcome is yes or no. What does that fact rule out?

- A. Random forests, which need numeric outcomes.
- B. Boosting, which only handles counts.
- C. Nothing, every model takes any outcome.
- D. Linear regression, which ignores the 0 to 1 bound.

Answer: D. The outcome type eliminates probability models one at a time, and it eliminates nothing on the other side, because trees, forests and boosting all have a classification mode. So when you decide against a forest, the reason is never the outcome. It is that somebody needs to read a number out of the model.

6. A lasso keeps 12 of 200 variables. You report their coefficients and p-values. What is wrong?

- A. Nothing, the lasso reports valid standard errors.
- B. They were shrunk, and chosen by looking at the data.
- C. Ridge would have been the right choice here.
- D. Twelve survivors is too few to mean anything.

Answer: B. Two separate things break at once. The penalty biases every coefficient toward zero by design, so the printed numbers are not estimating what they appear to. And these 12 won a 200-way search, while the inference is computed as though the model had been settled in advance. Treat what survives as a shortlist worth investigating.

7. A regulator will review every loan rejection. What does that alone tell you?

- A. You need a probability model.
- B. You need as much training data as possible.
- C. Accuracy is what matters, so pick the strongest model.
- D. A neural network will handle the complex rules.

Answer: A. Somebody is going to read a coefficient and ask you to defend it, and only a probability model has one to give. This was settled before anyone looked at a row of data. A boosted model might predict default better and still be unusable, because the algorithm said so is not a defence.

8. 300 rows, 80 candidate columns. What do you rule out straight away?
- A. Cross-validation, since the folds get too small.
 - B. Nothing, all models work at any size.
 - C. Forests and boosting, which you cannot afford here.
 - D. Linear regression, which cannot take 80 columns.

Answer: C. Flexibility is bought with sample size. At 300 rows and 80 columns a flexible model will fit beautifully and predict badly, because most of what it is fitting is noise. Regularized regression is the sensible move, used to cut 80 columns down to a few chosen for a reason.

9. Your forest beats your regression by five points of cross-validated error. Now what?
- A. Ship the forest, it won.
 - B. Average the two together.
 - C. Ship the regression, simpler is better.
 - D. Find out what the forest is using.

Answer: D. A gap that size is information, not a verdict. Usually the forest has found an interaction or a threshold you can name, and once you put it into the regression the gap closes and you ship the model you can explain. If it does not close, you have learned the structure is genuinely complicated, which is also worth knowing.

10. A depth-10 tree scores 0.44 training error and 13.19 cross-validated. A line scores 11.59 and 12.23. Which is better?
- A. The tree, since 0.44 is the lowest number here.
 - B. The line, by the column that counts.
 - C. Neither, they cannot be compared.
 - D. The tree, since it has more flexibility to use.

Answer: B. Training error measures how well a model reproduces rows it has already seen, and a flexible model drives that toward zero by memorizing. The honest column is the other one, and there the tree loses to a straight line. Both models sat on the same rows and were scored the same way, which is what lets you compare them at all.

11. A revenue model reports a cross-validated MSE of 140. What do you tell the owner?
- A. It explains 140 percent of the variation.
 - B. It is off by roughly 140 dollars on a typical day.
 - C. MSE does not translate into dollars.
 - D. It is off by roughly 12 dollars on a typical day.

Answer: D. MSE is in squared units, so 140 means 140 squared dollars, which nobody can picture. The square root of 140 is about 12, and that is back in dollars and is a sentence the owner can act on. Quote the RMSE for people, and keep the MSE for ranking models against each other.

12. A model that must run nightly for two years, with any output explainable on request. What should that push you toward?
- A. Whatever scored best in testing.
 - B. The most flexible model you can afford.
 - C. The simplest model that clears the bar.
 - D. A model retuned every month.

Answer: C. Accuracy is one requirement among several, and this job is mostly the other ones. A model with six tuned settings is six things that can drift as the data moves, and every one of them is something you would have to explain. Choosing simplicity on purpose reads as judgement. Defaulting to it because you never tried anything else does not.